



Demystifying Server Security:

A Beginner-Friendly Intro to Wazuh Docker and
openSUSE Leap 16



MATERIAL COMPILED BY



DARIAN RIZALUDIN
PT. RADNET DIGITAL INDONESIA
PART OF @KLAS_ACTIVITY
LINKEDIN @DARIANRIZALUDIN

PROJECT ASSISTANT

**AZARYA
YANUAR K**



**LOUIS
FACHRI**



**ANDREAN
WIJAYA**



THE PROBLEM: “IS MY SERVER ACTUALLY SAFE?”

A server can be running normally while being actively attacked.

server



🔒 SSH login attempts

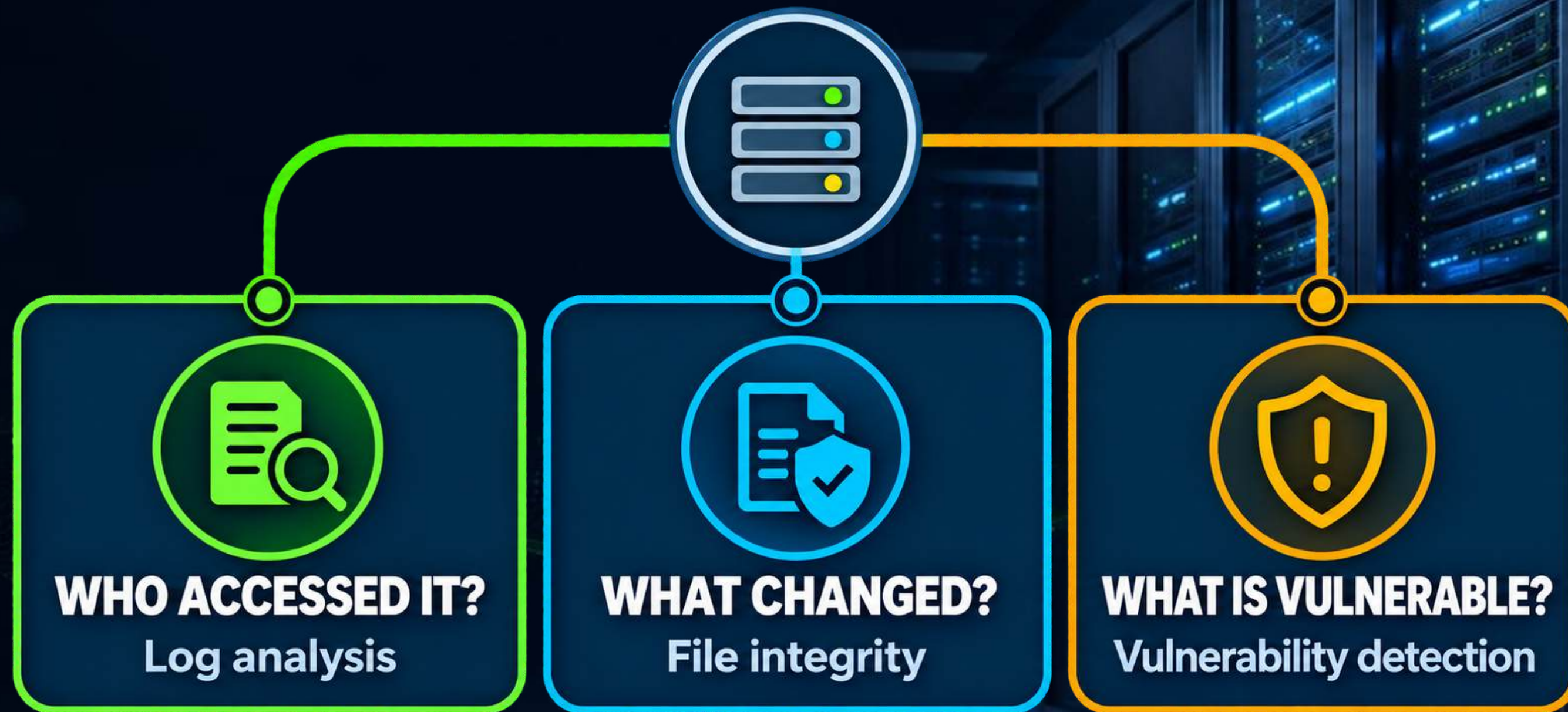
⚙️ Configuration changes

📦 New/updated packages

🕸️ Suspicious processes

⚠️ Security vulnerabilities

WHY SERVER SECURITY FEELS COMPLICATED



WHAT ARE WE ACTUALLY MONITORING ?



FROM “LOGS” TO “SECURITY INFORMATION”



WHAT IS WAZUH ?



WAZUH : THE BIG PICTURE



WHY OPENSUSE LEAP 16 ?



WHY DOCKER FOR WAZUH ?



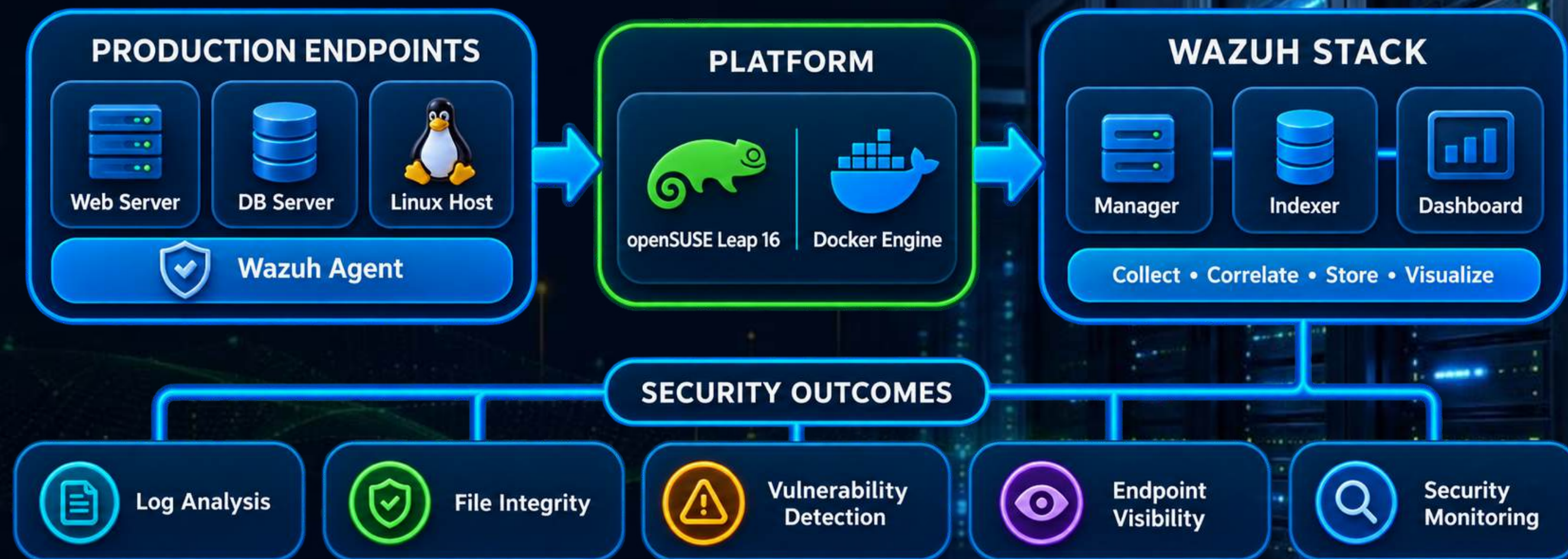
IMPORTANT

Docker simplifies deployment — not security.

Configuration • Updates • Good practices still matter



ARCHITECTURE



DEPLOYMENT JOURNEY



Security monitoring is a pipeline — not just an installation.



LOG ANALYSIS

Aug 14, 2026 @ 11:18:18.402 - Sep 14, 2026 @ 11:18:18.402					
Export Formatted Reset view 1247 available fields Columns Density 1 fields sorted Full screen					
	timestamp	agent.name	rule.description	rule.level	rule.id
	Sep 14, 2026 @ 11:18:02.7...		Auditd: SELinux permission check.	3	80730
	Sep 14, 2026 @ 11:18:02.7...		Auditd: SELinux permission check.	3	80730
	Sep 14, 2026 @ 11:17:42.8...		sshd: authentication failed.	5	5760
	Sep 14, 2026 @ 11:17:40.8...		PAM: User login failed.	5	5503
	Sep 14, 2026 @ 11:17:40.8...		unix_chkpwd: Password check failed.	5	5557
	Sep 14, 2026 @ 11:17:38.8...		sshd: authentication failed.	5	5760
	Sep 14, 2026 @ 11:17:38.8...		PAM: User login failed.	5	5503
	Sep 14, 2026 @ 11:17:38.8...		unix_chkpwd: Password check failed.	5	5557
	Sep 14, 2026 @ 11:17:22.8...		sshd: Attempt to login using a non-existent user	5	5710
	Sep 14, 2026 @ 11:17:20.8...		PAM: User login failed.	5	5503
	Sep 14, 2026 @ 11:17:20.8...		sshd: Attempt to login using a non-existent user	5	5710
	Sep 14, 2026 @ 11:17:19.5...		Integrity checksum changed.	7	550
	Sep 14, 2026 @ 11:17:19.5...		Integrity checksum changed.	7	550
	Sep 14, 2026 @ 11:17:19.5...		Integrity checksum changed.	7	550

Who is trying to access my server?



LOG ANALYSIS

- Username
- Source IP
- Event Description
- Timestamp

t	GeoLocation.city_name	Paris
t	GeoLocation.country_name	France
🌐	GeoLocation.location	{ "lon": 2.3527, "lat": 48.8543 }
t	GeoLocation.region_name	Paris
t	_index	wazuh-alerts-4.x-2026.09.14
t	agent.id	011
t	agent.ip	
t	agent.labels.os_group	linux
t	agent.name	
t	data.srcip	80.94.92.176
t	data.srcuser	lh
t	decoder.name	sshd
t	decoder.parent	sshd
t	full_log	Sep 14 03:22:06 sshd-session[2458114]: Failed password for inva lh from 80.94.92.176 port ssh2
t	id	1789356128.3017233
t	input.type	log
t	location	journald
t	manager.name	
t	predecoder.hostname	
t	predecoder.program_name	sshd-session
t	predecoder.timestamp	Sep 14 03:22:06
t	rule.description	sshd: Attempt to login using a non-existent user
#	rule.firedtimes	194
t	rule.gdpr	IV_35.7.d, IV_32.2
t	rule.gpg13	7.1
t	rule.groups	syslog, sshd, authentication_failed, invalid_login
t	rule.hipaa	164.312.b
t	rule_id	5710



LOG ANALYSIS

Weekly Notification Report

- What's Top IP
- Number of attempts

Detection ≠ automatic
proof of compromise.

 **UNKNOWN / NON-EXISTENT USERS**
Missing target-user field : 42,266
Rule 5710 (non-existent) : 42,200

 **SUCCESSFUL LOGIN SOURCES**
Only source IPs with at least 1 successful SSH authentication are listed.
1. [REDACTED] — Success: 5 | Failed: 0
Users: [REDACTED] | Target users: 1 |  **REVIEW**
2. [REDACTED] — Success: 1 | Failed: 0
Users: [REDACTED] | Target users: 1 |  **REVIEW**

 **THREAT HUNTING FINDINGS**

- High-volume SSH authentication failures detected (48,429 failed events).
- Successful SSH authentication observed (6 event(s)); investigate source IP and account.
- Non-existent-user SSH attempts (rule 5710): 42,200 event(s).
- 42,266 SSH events lacked a target username field; review rule 5710/non-existent-user activity.
- 10 Top source IP(s) targeted 3+ distinct accounts.
- Highest-volume source was [80.94.92.176](#) with 4,712 SSH events.
- Average SSH event volume per source IP was 1213, indicating concentrated activity.

 **MITRE ATT&CK CONTEXT**

- T1110.001 — Password Guessing
- T1021.004 — SSH

16:12

WazuhAlert_VPS
SSH THREAT HUNTING REPORT

 **Period:** 07 Sep 2026 01:00 → 14 Sep 2026 01:00
 **Agent:** [REDACTED]

 **OVERVIEW**
SSH events : 99,976
Failed attempts : 99,862
Successful logins : 7
Unique source IPs : 46
Targeted users : 33
Failure rate : 99.89%

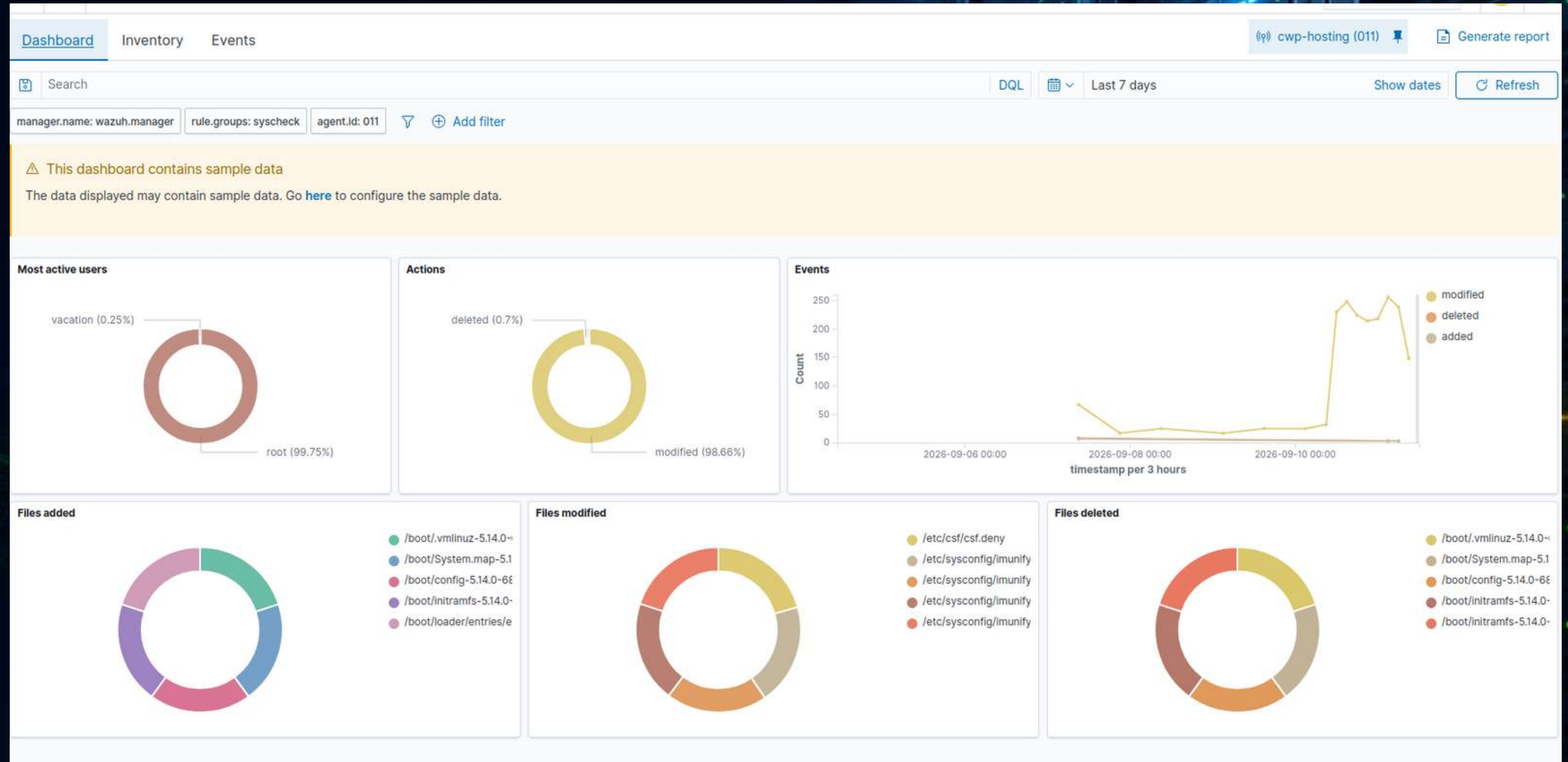
 **TOP 10 SOURCE IPs**

- [80.94.92.176](#)
Attempts: 10,224 | Failed: 10,224 | Success: 0
Target users: 11 |  **HIGH**
- [80.94.92.27](#)
Attempts: 9,986 | Failed: 9,986 | Success: 0
Target users: 13 |  **HIGH**
- [80.94.92.163](#)
Attempts: 9,976 | Failed: 9,976 | Success: 0
Target users: 11 |  **HIGH**
- [2.57.122.245](#)
Attempts: 9,972 | Failed: 9,972 | Success: 0
Target users: 8 |  **HIGH**
- [2.57.122.137](#)
Attempts: 9,943 | Failed: 9,943 | Success: 0
Target users: 9 |  **HIGH**
- [80.94.92.28](#)
Attempts: 9,941 | Failed: 9,941 | Success: 0
Target users: 12 |  **HIGH**
- [80.94.92.29](#)
Attempts: 9,271 | Failed: 9,271 | Success: 0
Target users: 12 |  **HIGH**
- [92.118.39.81](#)
Attempts: 9,002 | Failed: 9,002 | Success: 0
Target users: 10 |  **HIGH**
- [80.94.92.30](#)
Attempts: 8,950 | Failed: 8,950 | Success: 0
Target users: 12 |  **HIGH**
- [80.94.92.31](#)
Attempts: 8,073 | Failed: 8,073 | Success: 0
Target users: 9 |  **HIGH**



FILE INTEGRITY MONITORING

What changed
on my server?



FILE INTEGRITY MONITORING

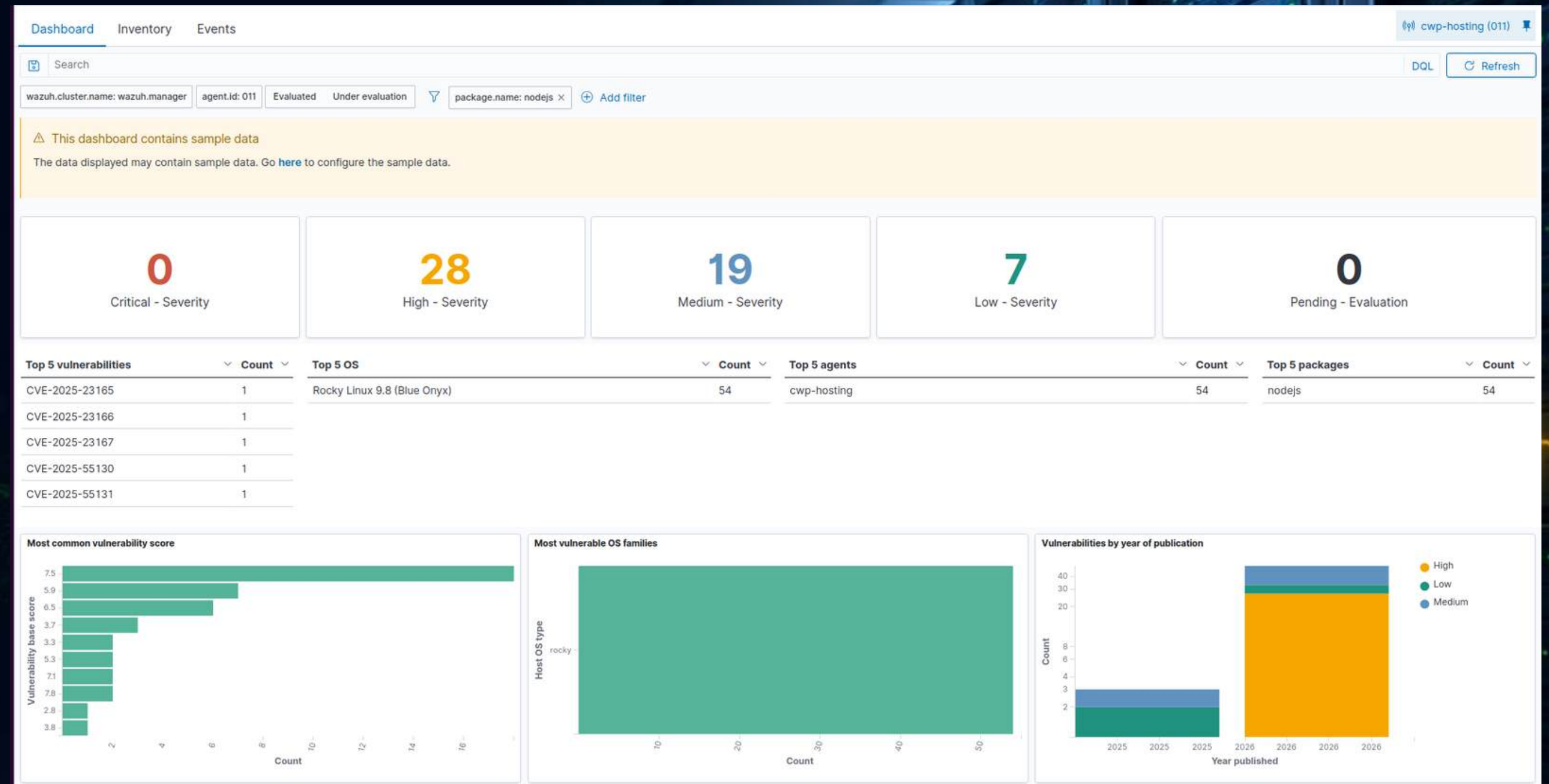
Detection ≠ automatic
proof of compromise.

Document Details		View surrounding documents	View single document
t _index	wazuh-alerts-4.x-2026.09.14		
t agent.id	011		
t agent.ip			
t agent.labels.os_group	linux		
t agent.name			
t decoder.name	syscheck_integrity_changed		
t full_log	File '/etc/sysconfig/immunify360/malware-filters-admin-conf/processed/ignored/av-admin-paths.txt' modified Mode: realtime Changed attributes: inode,mtime Old modification time was: '1789357635', now it is '1789358527' Old inode was: '9514', now it is '201332389'		
t id	1789358527.3532923		
t input.type	log		
t location	syscheck		
t manager.name	wazuh.manager		
t rule.description	Integrity checksum changed.		
# rule.firedtimes	16		
t rule.gdpr	II_5.1.f		
t rule.gpg13	4.11		
t rule.groups	ossec, syscheck, syscheck_entry_modified, syscheck_file		
t rule.hipaa	164.312.c.1, 164.312.c.2		
t rule.id	550		



VULNERABILITY DETECTION

What software
on my server is
vulnerable?



VULNERABILITY DETECTION

	agent.name	package.name	package.version	vulnerability.description	vulnerability.severity	vulnerability.id
		nodejs	1:16.20.2-8.el9_4	undici: undici: Man-in-th...	High	CVE-2026-9697
		nodejs	1:16.20.2-8.el9_4	undici: Undici: Information discl...	Medium	CVE-2026-9678
		nodejs	1:16.20.2-8.el9_4	ip-address: ip-address: Inconsi...	High	CVE-2026-69192
		nodejs	1:16.20.2-8.el9_4	undici: Undici: Response queue...	Low	CVE-2026-6733
		nodejs	1:16.20.2-8.el9_4	nodejs: Node.js: Permission by...	Low	CVE-2026-21716
		nodejs	1:16.20.2-8.el9_4	brace-expansion: brace-expan...	High	CVE-2026-69152
		nodejs	1:16.20.2-8.el9_4	nodejs: Node.js: Information dis...	Medium	CVE-2026-48615
		nodejs	1:16.20.2-8.el9_4	undici: undici: Weakening of co...	Low	CVE-2026-11525
		nodejs	1:16.20.2-8.el9_4	undici: undici: Denial of Service...	High	CVE-2026-1526
		nodejs	1:16.20.2-8.el9_4	undici: Undici: HTTP Request S...	High	CVE-2026-1525
		nodejs	1:16.20.2-8.el9_4	brace-expansion: Brace-expan...	High	CVE-2026-14257
		nodejs	1:16.20.2-8.el9_4	Node.js: Node.js: Unauthorized...	Medium	CVE-2026-21711
		nodejs	1:16.20.2-8.el9_4	brace-expansion: brace-expan...	Medium	CVE-2026-25547

Vulnerability management is a continuous cycle, not a one-time scan.

VULNERABILITY DETECTION

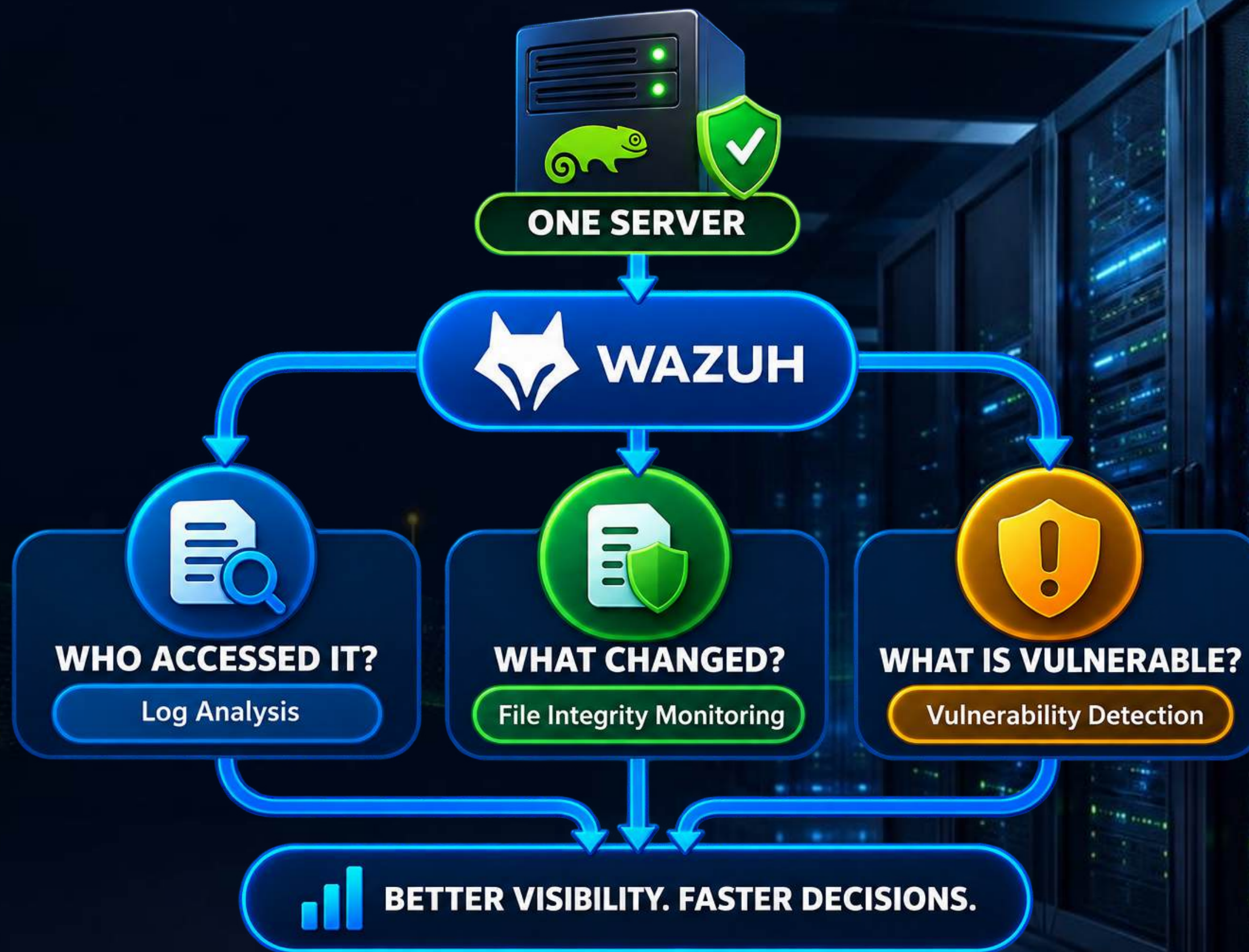
- Package name
- Package version
- Vulnerability description
- Vulnerability reference

Vulnerability details

Table JSON

_index	wazuh-states-vulnerabilities-wazuh.manager
agent.id	011
agent.name	
agent.type	Wazuh
agent.version	v4.14.6
host.os.full	
host.os.kernel	
host.os.name	
host.os.platform	
host.os.type	
host.os.version	
package.architecture	x86_64
package.description	JavaScript runtime
package.name	nodejs
package.size	378,133
package.type	rpm
package.version	1:16.20.2-8.el9_4
vulnerability.category	Packages
vulnerability.classification	-
vulnerability.description	undici: undici: Man-in-the-Middle attack via ignored TLS options with SOCKS5 proxy (CVE-2026-9697)
vulnerability.detected_at	Sep 12, 2026 @ 02:46:31.238
vulnerability.enumeration	CVE
vulnerability.id	CVE-2026-9697
vulnerability.published_at	Jun 18, 2026 @ 01:18:06.000
vulnerability.reference	https://bugzilla.redhat.com/show_bug.cgi?id=2476810 , https://bugzilla.redhat.com/show_bug.cgi?id=2489980 , https://bugzilla.redhat.com/show_bug.cgi?id=2490000 , https://bugzilla.redhat.com/show_bug.cgi?id=2490006 , https://bugzilla.redhat.com/show_bug.cgi?id=2490008 , https://bugzilla.redhat.com/show_bug.cgi?id=2490018 , https://bugzilla.redhat.com/show_bug.cgi?id=2490024 , https://bugzilla.redhat.com/show_bug.cgi?id=2493325 , https://bugzilla.redhat.com/show_bug.cgi?id=2493326 , https://bugzilla.redhat.com/show_bug.cgi?id=2493329 , https://bugzilla.redhat.com/show_bug.cgi?id=2493331 , https://bugzilla.redhat.com/show_bug.cgi?id=2493332 , https://bugzilla.redhat.com/show_bug.cgi?id=2493333 , https://bugzilla.redhat.com/show_bug.cgi?id=2493335 , https://bugzilla.redhat.com/show_bug.cgi?id=2493337
vulnerability.scanner.condition	Package less than 1:24.18.0-1.module+el9.8.0+40241+ba7fe110
vulnerability.scanner.reference	https://cti.wazuh.com/vulnerabilities/cves/CVE-2026-9697





WHAT WE LEARNED AT PT RADNET DIGITAL INDONESIA





QUESTIONS?

Your server is already generating security events.

ARE YOU LISTENING?



Source :

- <https://github.com/hanafiandrean/wazuh-opensuse.git>
- <https://github.com/Iszz100/Wazuh-all-in-one-yml.git>

Reference :

<https://documentation.wazuh.com/current/quickstart.html>

